# PARKING SYSTEM

OBJECT-ORIENTED PROGRAMMING

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# DESCRIPTION OF THE SYSTEM

## Purpose

The purpose of this system is to manage parking slots in an organized and efficient way. It helps users reserve parking spaces, park vehicles, track parking duration, and calculate payment based on slot type and time.

## Main Features

- Show all parking slots
- Reserve a parking slot
- Park a vehicle
- Remove a vehicle
- Calculate payment
- Show reservations
- Show payment records
- Detect if a vehicle parks in a different slot from its reservation

# LIST OF MAIN CLASSES

01. Vehicle
02. ParkingSlot
03. SunnySlot
04. ShadedSlot
05. Reservation
06. Payment
07. ParkingArea
08. ParkingException

# RESPONSIBILITY OF EACH CLASS

### Vehicle

Stores vehicle information such as vehicle ID and owner name.

### ParkingSlot

Represents a parking slot and stores slot number, parking status, parked vehicle, and parking start hour.

### SunnySlot

A type of parking slot with a lower hourly rate.

### ShadedSlot

A type of parking slot with a higher hourly rate.

### Reservation

Stores reservation details such as vehicle ID and reserved slot number.

### Payment

Calculates the total parking payment using duration, slot rate, and penalty if needed.

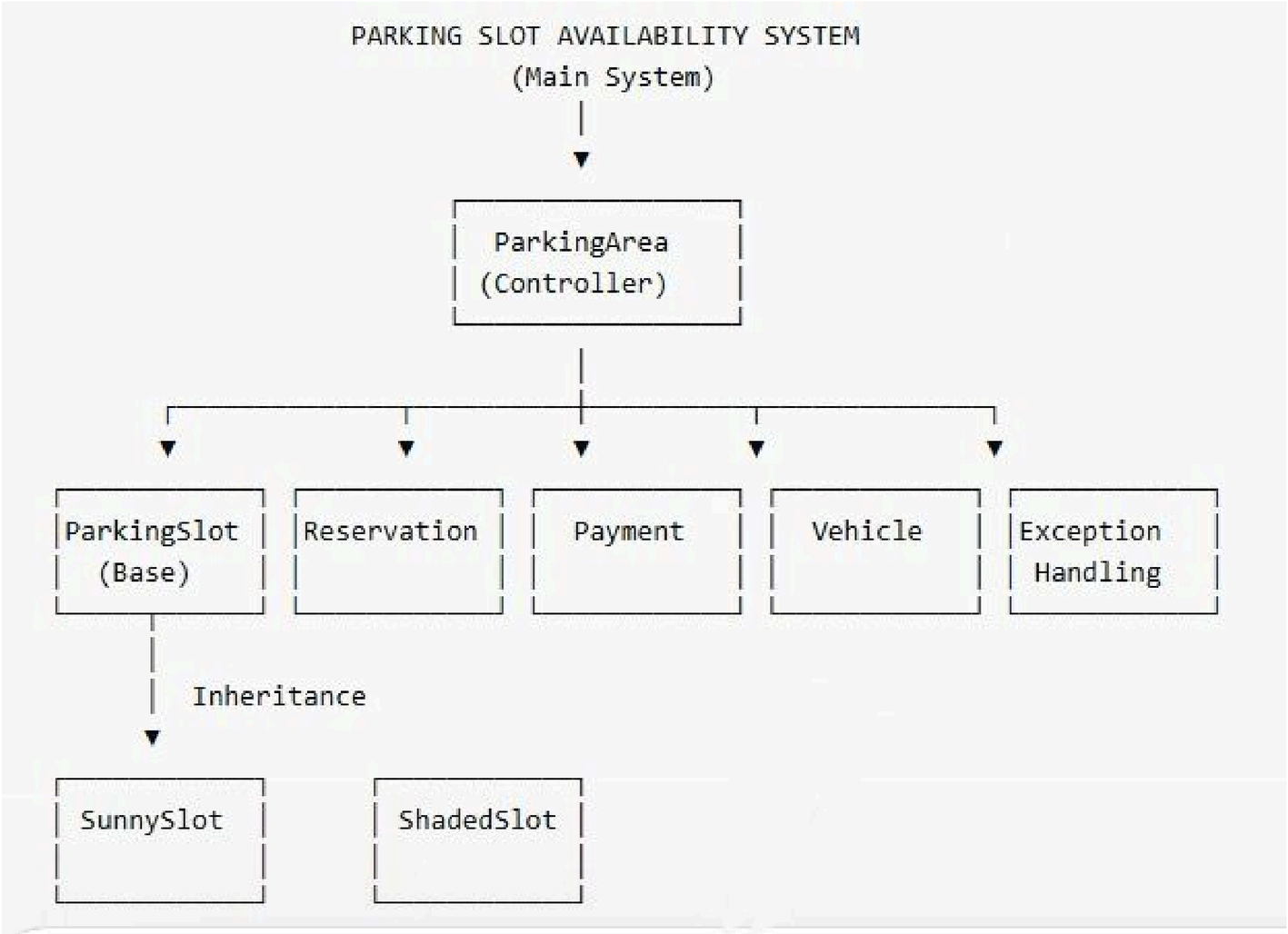
### ParkingArea

Manages all slots, reservations, payments, and system operations.

### ParkingException

Handles errors in the system, such as invalid slot number or vehicle not found.

# RELATIONSHIPS BETWEEN CLASSES

| Inheritance | Composition |
| --- | --- |
| • SunnySlot inherits from ParkingSlot<br>• ShadedSlot inherits from ParkingSlot | • ParkingArea has many ParkingSlot objects<br>• ParkingArea has many Reservation objects<br>• ParkingArea has many Payment objects |

# CLASSES VISUALIZATIOLN